

AMBIENT BACKSCATTER: Power And Communication To Battery-Free Devices



Dr Rajiv Kumar Singh

holds a PhD in electronics engineering from IIT (BHU), Varanasi, and is currently employed with Institute of Engineering & Technology (IET), Lucknow as assistant professor. He has received senior research fellowship of UGC at Centre of Advanced Study at IIT, Varanasi. His interests include wired and wireless technologies for high-speed Internet access

Chandan Kumar

Maurya is pursuing BTech in electronics and instrumentation engineering from IET, Lucknow

With recent advancements in communication technology, there is a new technique—called ambient backscatter—that allows devices to communicate with each other, for which it does not require batteries. Let us take a glimpse at what ambient backscatter is

Ambient backscatter is a new type of wireless communication that uses ambient signals in radio frequency (RF) range to enable communication between devices and powering these without using batteries. This system utilises existing ambient RF radiations as the source of power and for communication between devices. It makes use of ambient RF signals such as TV broadcast signals, mobile communication signals, Wi-Fi and so on. For all such applications, ambient backscatter does not require a separate power source to operate.

An ambient backscatter device communicates by absorbing and reflecting RF signals. In a backscatter communication system, a device communicates by modulating the reflection of incident RF signals (instead of generating RF signals). Hence, communication using this technique is more energy efficient than conventional radio communication.

How it works

The working of ambient backscatter communication can be understood by the following example. Consider two nearby battery-free devices (A and B) and a TV tower as source of ambient RF signals in a certain area with good signal strength, as shown in Fig. 1. Suppose, A wants to send a message to B. To do so, A backscatters ambient signals to convey the message in the form of bits—it can be indicated by switching the device's antenna between reflecting and non-reflecting states corresponding to either bit 0 or 1.

Signals that are reflected by A create an additional path from the TV tower to B. On the other hand, B detects the change in signals caused by backscattering and decodes A's message using some decoder logic.

General architecture of ambient backscatter

The basic design of ambient backscatter consists of mainly three parts: harvester, transmitter and receiver (Fig. 2). Harvester is the circuitry responsible for the extraction of energy from ambient RF signals to provide power to the battery-free device.

Transmitter conveys the message by backscattering RF signals that are incident on its antenna by modulating the antenna to reflect or absorb signals.

On the receiver, an antenna intercepts some of the power of RF signals backscattered by the transmitter. Finally, the message is decoded by using a decoding technique.

RF energy harvester. An RF energy harvester module consists of an antenna that receives ambient RF signals. Received signals are amplified by voltage doubler/multiplier, which does not require any

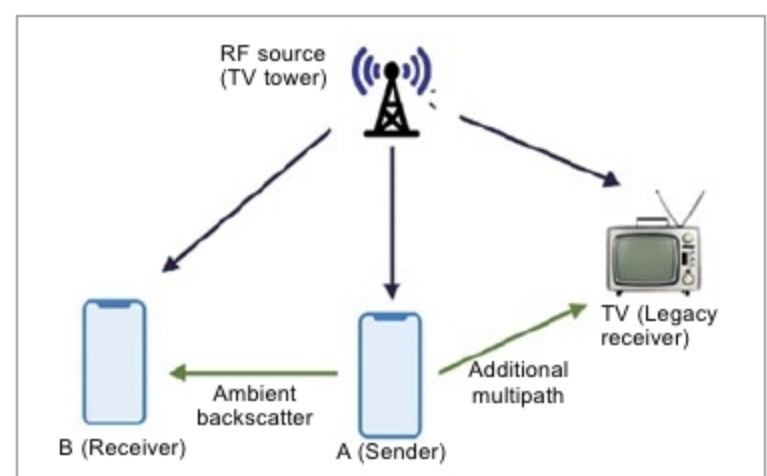


Fig. 1: Communication between two battery-free devices using ambient backscatter

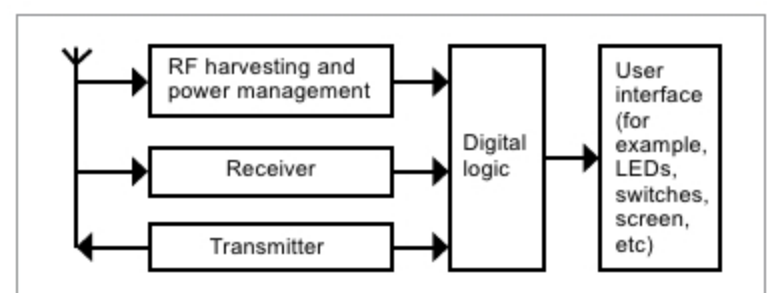


Fig. 2: Block diagram of ambient backscattering device